

RESEARCH

Open Access



Using the multilayer perceptron approach to explore the relationship between PUBG gaming, sleep disorder, quality of life, and migraine

Ali Hassan¹, Muhammad Daniyal^{2*}, Roy Rillera Marzo³, Mohammed Aljuaid⁴ and Duaa Shahid⁵

Abstract

Background of the study Player Unknown's Battlegrounds (PUBG), a popular and widely played multiplayer online game, has generated interest and concern about its effects on the physical health of its players. This study explores the relationship between factors like cultivation level, gaming disorder, migraine and associated symptoms, sleep quality, and life quality of PUBG players.

Methods This cross-sectional study included 500 PUBG players, categorized into Lower, Medium, and High PUBG Users. Data were collected using a self-administered questionnaire, including the Pittsburgh Sleep Quality Index (PSQI), Epworth Sleepiness Scale (ESS), and a gaming disorder screening tool. The Multilayer Perceptron (MLP) methodology was applied to analyze the factors influencing migraine symptoms, sleep quality, and quality of life.

Results Participants of the study were categorized into lower PUBG users (LPU), medium PUBG users (MPU), and high PUBG users (HPU). Among study participants reporting migraine pain, 259 (51.8%) reported that they were HPU. By examining daytime sleepiness using the ESS, Higher normal Day sleep (DS) was observed in 78 (15.9%) HPU. The cultivation level of PUBG showed a very weak positive correlation with experiencing migraine pain or associated headache symptoms ($r=0.034$, $p=0.454$). In contrast, the gaming disorder of PUBG showed a weak negative correlation with PSQI ($r=-0.092$, $p=0.041$). The higher levels of gaming disorder are slightly associated with poorer sleep quality. The results of the MLP model suggested that daily PUBG use was the most contributing factor to migraine and related symptoms followed by gaming disorder, gaming addiction, PSQI, and ESS.

Conclusion The study concluded that PUBG playing contributes to migraine and its associated symptoms although is not significant it contributes to less sleep quality and lower quality of life.

Keywords Migraine disorders, Sleep quality, Gaming disorder, Daytime sleepiness, Machine learning

*Correspondence:

Muhammad Daniyal
muhammad.daniyal@iub.edu.pk;
muhammad.daniyal@almoosahealth.com.sa

¹Department of Media and Communication Studies, The Islamia University of Bahawalpur, Bahawalpur, Pakistan

²Research Center, Almoosa Specialist Hospital, Al-Ahsa, Saudi Arabia

³Faculty of Humanities and Health Sciences, Curtin University, Miri, Sarawak, Malaysia

⁴Department of Health Administration, College of Business Administration, King Saud University, Riyadh, Saudi Arabia

⁵Hult International Business School, Cambridge, MA 02141, USA



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Advancements in technology have significantly increased the use of electronic gadgets for leisure, impacting adolescents' daily lives and consuming substantial time [1]. Devices such as computers, tablets, TVs, electronic games, and cell phones are used for both educational and recreational purposes [2]. Easy access to electronics has led to increased time spent using them, which has been linked to detrimental effects on adolescents' health that have been documented in the literature. These effects include altered sleep patterns, exhaustion, anxiety, depression, obesity, a decrease in physical activity, a decrease in concentration, musculoskeletal pain, stress, and headaches [3–7]. Some studies have indicated that headache is a major complaint among adolescents due to the excessive use of electronic devices. It has been pointed out as an emerging concern among children and adolescents and may be related to changes in lifestyle and stress-related factors [6, 8, 9]. Moreover, bruxism is a significant factor affecting sleep and may be related to headaches [10, 11]. Similarly, there is a 70% increased risk of headaches for people who use computers for work or play for longer than three hours. Online video gaming has converted into one of the world's most widespread leisure time activities among adolescents [12]. Numerous beneficial effects of gaming have been repeatedly demonstrated by research, including therapeutic, physiological, medical, cognitive, and educational advantages [13]. Primary headaches are painful and often debilitating without an underlying cause [14]. The International Classification of Headache Disorders (ICHD-3) identifies common types as migraines and tension-type headaches, with migraines causing throbbing pain, nausea, and light sensitivity, and tension-type headaches causing a tight, pressure-like sensation [15]. Understanding these headaches is essential for accurate diagnosis and treatment, as they significantly affect quality of life, academic performance, and social interactions, especially in children and adolescents who are undergoing rapid developmental changes [16–19]. Increasing concern over gaming addiction highlights its impact on mental health and family relationships, with studies in adults suggesting a link between screen exposure and headaches [20, 21]. Additionally, it has been found that engaging in online gaming can lead to tension-type headaches [22]. In a similar vein, a study including adult participants revealed a correlation between the use of smartphones and the occurrence of new-onset main headaches [23]. Research on the impact of online games on pediatric headaches is currently limited. However, during the COVID-19 pandemic, studies have explored the correlation between internet addiction, smartphone use, and headaches. One such study assessed children and adolescents diagnosed with tension-type headaches or migraines, evaluating their levels

of smartphone and online game addiction and comparing them to their peers. Additionally, these children were referred to the Child Psychiatry department to investigate potential correlations between psychiatric illnesses, if present, and the occurrence of headaches or smartphone addiction [24, 25]. In the United States, 6% of children aged 5–17 experience frequent or severe headaches, impacting their daily activities and quality of life [26–28]. Pediatric migraines can significantly affect academic and emotional well-being, sometimes more severely than chronic illnesses like rheumatic diseases and cancer [26]. Migraine development is linked to bright and complex visual stimuli, and screen exposure is commonly reported to trigger or worsen headaches [29]. A study of 320 children in Italy found that 30.7% with migraine with aura, 22.6% with migraine without aura, and 3.8% with tension-type headaches reported headaches from screen use [30]. Additionally, video gaming was the fourth most common trigger in a study of 102 migraine patients, and extended screen time, including phone use, is associated with increased headache frequency [31–33]. Subsequent research has produced contradictory findings about the relationship between screen exposure and the occurrence of headaches. A cross-sectional study was conducted in Finland to assess the relationships between various types of headaches and leisure activities among 13-year-olds in the population. The study had a total of 59 children diagnosed with migraine, 65 children diagnosed with episodic tension-type headaches, and 59 control participants who were free of headaches. Although the specific type of leisure activity did not show a significant correlation with the type of headache, it was observed that children with migraine and episodic tension-type headaches tended to use computers more frequently compared to children without headaches. Additional analysis revealed that using a computer for one extra day per week significantly increased the likelihood of being in the migraine group (with a 20% increase) and the tension-type group (with a 30% increase), compared to the control group [22]. Similarly, a population-based study conducted in Saudi Arabia examined the correlation between the utilization of various forms of electronic media (such as mobile phones, computers, gaming consoles, television, and music listening) and the occurrence of headaches among 1025 adolescents aged 13–17 years. The study indicated that the occurrence of any form of headache rose as the time of daily music listening increased. However, no other statistically significant connections were established between the usage of electronic media and various types of headaches [16]. A recent study in France found that high screen time is associated with an increased risk of migraines in young adults, but not with non-migraine headaches [8]. PUBG, popular among adolescents for its competitive gameplay [34], is linked to gaming addiction

due to the significant developmental changes during adolescence [35]. While research on PUBG's impact on physical and psychological well-being is extensive, including issues like reduced leisure activity, poorer sleep, and lower life satisfaction [36] evidence connecting PUBG to migraines in children remains limited [37–39].

However, there is a general lack of knowledge in this field, particularly regarding how prolonged PUBG playing sessions may further affect the severity of migraine-related symptoms and disability. Considering the lack of information in the existing literature, we undertook our present study to further examine the correlation between teenage PUBG usage, specifically in terms of duration, and migraine. The study hypothesis posited that adolescents who reported the longest period of playing PUBG would experience a higher frequency of migraines, as well as increased absenteeism and reduced functioning at school or college.

Materials and methods

Study design and sample size calculation

This cross-sectional study investigated the relationship between PUBG gaming and its impact on migraine, internet gaming disorder, sleep quality, and overall life quality. The required sample size was estimated using the Raosoft sample size calculator. With a margin of error of $\alpha=5\%$ and a 95% C.I, and based on estimated population of Sargodha City of Pakistan the initial calculated sample size, $n=541$ respondents. Considering the potential for non-response and incomplete responses, the final target sample size was adjusted to 500 respondents. The data collection for this study was conducted between March 2022 and July 2022. To initiate the process, potential respondents were posed a straightforward question: "Do you play PUBG game on your cell phone?" Those individuals who responded affirmatively, indicating that they indeed engaged in playing PUBG on their mobile devices, were then selected to participate in the study. The reliability of the questionnaire was assessed using Cronbach's alpha, which is a typical way of checking the internal consistency of a test. The total reliability of the study was high, with a Cronbach's alpha (α) of 0.870. Specifically, the Cronbach's alpha values for the individual components were: GAS = 0.786, IGD = 0.880, ESS = 0.765, PSQI = 0.687, MQoL = 0.730, and cultivation level = 0.701. The internal consistency of the various scales and measures used in the study is good, as seen in these values. The response of the study mainly includes the qualitative variables including responses from the questionnaire. So chi-square test of independence has been applied to find the association between different qualitative attributes. Furthermore, spearman's rank correlation has been applied to measure the strength of association. The significance of the statistical tests has been measured by

p-value and effect size. For the computation purpose of effect size, you can use Cramér's V, which is commonly used for chi-square tests with more than 2×2 contingency tables.

The formula for Cramér's V is:

$$V = \sqrt{\frac{\chi^2}{N(k-1)}}$$

Here N is the total number of observations, k is the smallest number of rows and columns. A value of **0.1** indicates a small effect, **0.3** represents a medium effect, and **0.5** signifies a large effect.

Inclusion and exclusion criteria

The study included male and female respondents who played PUBG on their cell phones and were proficient enough to understand the questionnaire. The exclusion criteria consisted of individuals who did play PUBG but were excluded for other reasons like non response, incomplete questionnaire.

Study design

This cross-sectional study was conducted between March 2022 and July 2022, the research employs a cross-sectional survey design using a structured questionnaire administered through Google Forms. The target population is residents of Sargodha, a city in Pakistan, across various demographic segments and was approved by the Social and Health Welfare Department with Ethical Approval No.26/H-11-2021. The reliability of the questionnaire was evaluated by employing Cronbach's alpha, a statistical measure that assesses the internal consistency of the instrument. Descriptive statistics, such as mean age, and PUBG hours daily were used to describe the demographic and gaming disorder, gaming addiction, PSQI (Pittsburgh Sleep Quality Index), and MQoL (Migraine Quality of Life) of the participants. The association between qualitative variables has been computed by the chi-square test of associations and the strength of association was computed by Spearman's correlation coefficient. Machine learning methodology has been extensively used to measure the relationship between different qualitative variables. This methodology involves algorithms and statistical models enabling computers to learn from data and make predictions or decisions without being explicitly programmed. One of the latest technique of ML method, MLP explained in next section, will be applied to measure the relative importance of each predictor variable in evaluating the impact of PUBG gaming on the migraine level, sleep quality, and quality of life. The statistical analyses were conducted using SPSS version 27; $p < 0.05$ was considered significant.

Statistical analysis

The statistical analysis was conducted using multiple methods to examine the relationships between the study variables. A Chi-Square Test was performed to compare the distribution of migraine occurrence across different levels of PUBG usage (LPU, MPU, HPU). Additionally, the Effect Size was estimated to assess the strength of associations between the predictors (e.g., gaming addiction, smartphone use, and sleep quality) and migraine symptoms. The strength of the correlation between the variables was estimated using Pearson's Correlation. The Multilayer Perceptron (MLP) method, a machine learning technique, was applied to assess the impact of various predictors (e.g., gaming addiction, smartphone use, and sleep quality) on migraine symptoms, allowing for a more nuanced understanding of the data.

Study tool

The survey included several sections, beginning with the collection of demographic data such as gender, age, area of living, and educational level. Additionally, the survey assessed the time spent playing PUBG on mobile devices. Various scales were used to measure different variables.

The Gaming Addiction Scale (GAS)

Gaming addiction, recognized as a behavioral disorder, is characterized by excessive video game play that disturbs daily life and responsibilities. It includes symptoms such as preoccupation with gaming, inability to reduce playtime, and neglect of personal and professional obligations. This disorder has been increasingly identified in recent years, with prevalence rates rising due to the growing popularity of online and interactive gaming. The GAS was employed to evaluate the extent of problematic gaming behavior. It consisted of seven items, with higher scores reflecting greater levels of gaming addiction. Participants rated the frequency of their gaming activities on a five-point Likert scale, where 1 represented "never" and 5 indicated "very frequently." [40].

Internet gaming disorder (IGD)

Internet addiction is a behavioral disorder characterized by excessive use of the Internet, impacting daily life and responsibilities. It encompasses various forms, including social media addiction, compulsive online gaming, and excessive web browsing. Symptoms often involve neglect of real-world obligations, social withdrawal, and impaired functioning. The prevalence of internet addiction has surged with the proliferation of digital technology, affecting a broad demographic, including adolescents and adults. The IGD Scale 9—Short Form (IGDS9-SF) was utilized to measure the severity of IGD symptoms. This nine-item scale used a 5-point Likert scale, with options from 1 (never) to 5 (very often). Higher scores on the

scale indicated a greater level of problematic internet gaming behavior.

Epworth sleepiness scale (ESS)

The ESS has been designed to assess an individual's overall daytime sleepiness across eight different scenarios. It utilizes a four-point Likert scale, with scores ranging from 0 to 24. Based on the total score, daytime sleepiness is categorized from "lower normal daytime sleepiness" to "severe excessive daytime sleepiness" [41].

Pittsburgh Sleep Quality Index (PSQI)

The PSQI measures sleep duration, quality, and disorders. The total PSQI score is the sum of these component scores. A score above 5 indicates poor sleep quality. The validity and reliability of the PSQI were assessed by Buysse et al. [42].

Migraine Quality of Life(24-hour MQoL)

The 24-hour MQoL Questionnaire includes 15 items across three components: symptoms, life constraints, and the psychological and social impacts of migraines. A higher overall score indicates a better quality of life with the questionnaire's cultural validity and reliability [43].

Cultivation level of PUBG

Cultivation Theory, which was initially employed to analyze the impact of conventional media such as television, has been utilized to evaluate the cultivation effect of video gaming. The participants were classified into three tiers based on their level of engagement: low cultivation (playing for less than 2 h), medium cultivation (playing between 2 and 4 h), and high cultivation (playing for more than 4 h) [44].

Results

Out of 500 respondents, there were 412 males (82.4%) and 88 females (17.6%). The average age of the 500 participants was 20.20 years $\pm$ 2.01. The frequency of migraine occurrences was 1.37 times per month ($SD = 1.89$), and the average daily duration of PUBG use was notably high at 5.17 h ($SD = 1.75$), reflecting significant engagement with this mobile game. Education levels among participants varied, Pre-graduate level (150 participants, 30.2%), intermediate education level (46 participants, 9.3%), and education above graduate level (6 participants, 1.2%). Regarding the area of living, a majority of participants resided in urban areas (307 participants, 61.4%), compared to those from rural areas (193 participants, 38.6%) [Table 1].

Among participants reporting migraine pain, 41 (8.2%) were LPU, 138 (27.6%) were MPU, and 259 (51.8%) were HPU. For those without migraine complaints, the distribution was 4 (0.8%) LPU, 19 (3.8%) MPU, and 39 (7.8%)

Table 1 Distribution of study sample according their demographic characteristics (N = 500)

Demographic Variables	Mean ± S.D		
Age (years)	20.20 ± 2.01		
Frequency of migraine (monthly)	1.37 ± 1.89		
Daily PUBG use (hours)	5.17 ± 1.75		
		N	Percentage
Gender	Male	412	82.4%
	Female	88	17.6%
Education level	Intermediate	46	9.3%
	Secondary	105	21.1%
	Pre-graduate	150	30.2%
	Graduate	190	38.2%
	Post-Graduate	6	1.2%
Area of living	Urban	307	61.4%
	Rural	193	38.6%

N = number of participants

S.D = standard deviation

HPU. The association was not statistically significant (χ^2 -test-statistic = 0.653, $p = 0.721$). Regarding PUBG gaming addiction, low gaming addiction was found in 18 (3.6%) of LPU, 64 (12.8%) of MPU, and 129 (25.8%) of HPU, while high gaming addiction was present in 27 (5.4%) of LPU, 93 (18.6%) of MPU, and 169 (33.8%) of HPU. The association between gaming addiction and PUBG cultivation level was not statistically significant (χ^2 -statistic = 0.367, $p = 0.832$). In terms of PUBG gaming disorder, low gaming disorder was observed in 10 (2.0%) of LPU, 72 (14.4%) of MPU, and 134 (26.8%) of HPU, while high gaming disorder was seen in 35 (7.0%) of LPU, 85 (17.0%) of MPU and 164 (32.8%) of HPU. This association was statistically significant (χ^2 -statistic = 8.902, $p = 0.012^{***}$). Examining sleep quality using the PSQI, good sleep quality was reported by 42 (8.5%) of LPU, 132 (26.6%) of MPU, and 253 (50.9%) of HPU, while bad sleep quality was reported by 3 (0.6%) of LPU, 24 (4.8%) of MPU, and 43 (8.7%) of HPU. The association between sleep quality and PUBG cultivation level was not statistically significant (χ^2 -statistic = 2.312, $p = 0.315$). For quality of life as measured by the 24-hour MQoL, low quality of life was reported by 22 (4.5%) of LPU, 86 (17.5%) of MPU, and 142 (28.9%) of HPU, while high quality of life was observed in 20 (4.1%) of LPU, 71 (14.5%) of MPU, and 150 (30.5%) of HPU. The association between quality of life and PUBG cultivation level was not statistically significant (χ^2 -statistic = 1.583, $p = 0.453$). The relationship between PUBG cultivation and migraine complaints was minimal, with an effect size (Cramér's V) of 0.036, indicating a negligible difference in migraine occurrence across cultivation levels. Similarly, the association with gaming addiction levels was weak, with an effect size of 0.027, showing little variation in addiction across users. However, a more notable association was found with gaming disorder levels, where

the effect size was 0.133, indicating a moderate relationship. This suggests that higher PUBG cultivation is more strongly associated with increased gaming disorder levels.

In contrast, the association with sleep quality (PSQI) was small, with an effect size of 0.068, implying minimal differences in sleep quality between varying levels of PUBG cultivation. Likewise, the impact on quality of life (24-hmqol) was also small, with an effect size of 0.056, showing only slight variation in quality of life across different cultivation levels. Finally, the effect size for daytime sleep disorders (ESS) was 0.083, which suggests a small but noticeable association between PUBG cultivation levels and daytime sleep disturbances.

By examining daytime sleepiness using the ESS, lower normal daytime sleepiness (DS) was found in 23 (4.7%) of LPU, 82 (16.7%) of MPU, and 130 (26.5%) of HPU. Higher normal DS was observed in 8 (1.6%) of LPU, 35 (7.1%) of MPU, and 78 (15.9%) of HPU. Mild excessive DS was reported by 0 (0.0%) of LPU, 5 (1.0%) of MPU, and 13 (2.7%) of HPU. Significant associations were found between PUBG cultivation levels and both gaming disorder and daytime sleepiness as measured by ESS, suggesting that higher levels of PUBG usage are linked with more severe gaming disorders and increased daytime sleepiness. Moderate excessive DS was found in 6 (1.2%) of LPU, 14 (2.9%) of MPU, and 28 (5.7%) of HPU. Severe excessive DS was reported by 5 (1.0%) of LPU, 19 (3.9%) of MPU, and 44 (9.0%) of HPU. The association between daytime sleepiness and PUBG cultivation level was statistically significant (χ^2 -statistic = 6.997, $p = 0.045$). Migraine complaints, gaming addiction, sleep quality, and quality of life, did not show statistically significant associations with PUBG cultivation levels [Table 2].

Cultivation level of PUBG gaming showed a very weak positive correlation with experiencing migraine pain or associated headache symptoms ($r = 0.034$, $p = 0.454$), gaming addiction ($r = -0.026$, $p = 0.560$), PSQI ($r = 0.041$, $p = 0.362$), 24-hMqoLQ ($r = 0.046$, $p = 0.311$), and ESS ($r = 0.053$, $p = 0.244$), indicating no significant relationships. However, there is a weak negative correlation between cultivation level and gaming disorder ($r = -0.091$, $p = 0.043^{**}$), significant at the 0.05 level, suggesting that higher cultivation levels are slightly associated with lower gaming disorder symptoms. Experiencing migraine pain or associated headache symptoms has very weak negative correlations with gaming addiction ($r = -0.059$, $p = 0.185$), gaming disorder ($r = -0.076$, $p = 0.089$), 24-hMqoLQ ($r = -0.036$, $p = 0.422$), and very weak positive correlations with PSQI ($r = 0.022$, $p = 0.622$) and ESS ($r = 0.041$, $p = 0.366$), indicating no significant relationships. Gaming addiction shows a significant moderate positive correlation with gaming disorder ($r = 0.489$, $p < 0.001$), highlighting a strong relationship where higher

Table 2 Association between sleep, migraine, and quality of life variables with PUBG cultivation level (N = 500)

Variables	Categories	Cultivation level			χ^2 - statistic (p-value)	Ef- fect Size (d)
		Lower PUBG users (LPU)	Medium PUBG users (MPU)	High PUBG users (HPU)		
		N (%)	N (%)	N (%)		
Do you have migraine pain or any com- plaints regarding migraine?	Yes	41 (8.2%)	138(27.6%)	259(51.8%)	0.653(0.721)	0.036
	No	4(0.8%)	19 (3.8%)	39(7.8%)		
PUBG GA (Gaming Addiction) levels	Lower GA level	18(3.6%)	64(12.8%)	129(25.8%)	0.367 (0.832)	0.027
	High GA level	27(5.4%)	93(18.6%)	169(33.8%)		
PUBG GD (Gaming disorder) Levels	Low GD Level	10(2.0%)	72(14.4%)	134(26.8%)	8.902 (0.012***)	0.133
	High GD Level	35(7.0%)	85(17.0%)	164(32.8%)		
PSQI (Pittsburgh Sleep Quality Index)	Good SQ	42(8.5%)	132(26.6%)	253(50.9%)	2.312(0.315)	0.068
	Bad SQ	3(0.6%)	24 (4.8%)	43(8.7%)		
24-hour MQoL (Quality of Life)	Low QOL	22(4.5%)	86(17.5%)	142(28.9%)	1.583 (0.453)	0.056
	High QOL	20(4.1%)	71(14.5%)	150(30.5%)		
ESS (Epworth sleepiness scale)	Lower Normal DS	23(4.7%)	82(16.7%)	130(26.5%)	6.997 (0.045**)	0.083
	Higher Normal DS	8(1.6%)	35(7.1%)	78(15.9%)		
	Mild Excessive DS	0(0.0%)	5(1.0%)	13(2.7%)		
	Moderate Excessive DS	6(1.2%)	14(2.9%)	28(5.7%)		
	Severe Excessive DS	5(1.0%)	19(3.9%)	44(9.0%)		

GD = Gaming Disorder

GA = Gaming Addiction

SQ = Sleep Quality

DS = Daytime Sleep

*** Highly Significant

**Significant

p-value was calculated by χ^2 - test of association/independence χ^2 - statistic = chi-square test of association

gaming addiction is associated with higher levels of gaming disorder. Additionally, gaming addiction has a significantly weak negative correlation with quality of life ($r = -0.139$, $p = 0.002^{***}$), suggesting that higher gaming addiction is associated with a slight decrease in quality of life. There are very weak and non-significant correlations between gaming addiction and PSQI ($r = -0.005$, $p = 0.912$) and ESS ($r = 0.027$, $p = 0.550$). Gaming disorder exhibits a weak negative correlation with PSQI ($r = -0.092$, $p = 0.041^{**}$), significant at the 5% level, indicating that higher levels of gaming disorder are slightly associated with poorer sleep quality. There are very weak and non-significant correlations between gaming disorder and 24-hour MQoL ($r = -0.049$, $p = 0.279$) and ESS ($r = -0.027$, $p = 0.551$). PSQI has very weak and non-significant correlations with 24-hMQoLQ ($r = -0.038$, $p = 0.397$) and ESS ($r = 0.026$, $p = 0.572$). Similarly, 24-hMQoLQ shows a very weak and non-significant correlation with ESS ($r = 0.017$, $p = 0.707$) [Fig. 1].

Machine learning approach for estimating the important predictors of migraine level

A neural network model is a supervised machine learning approach used to estimate the complex relationships between dependent and independent variables.

This model operates through a network of interconnected nodes, or neurons, organized into layers. Neural networks are highly versatile and can handle various tasks, including regression, where continuous values are predicted and classification, where data is categorized into distinct labels, is performed. The network typically includes:

1. **Input Layer:** Receives the initial data.
2. **Hidden Layers:** Transform the input data through weighted connections and activation functions.
3. **Output Layer:** Produces the final prediction or classification.

Neural networks are versatile and can be used for tasks like regression (predicting continuous values) and classification (assigning labels to data).

In our study, the neural network model will be employed to investigate the impact of several factors on migraine pain and associated headache symptoms. Specifically, you will analyze how variables such as PUBG cultivation level, gaming addiction, gaming disorder, and indicators of sleep quality (ESS), quality of life (MQoL), and sleep problems (PSQI) influence the experience of migraines. By incorporating these independent variables,

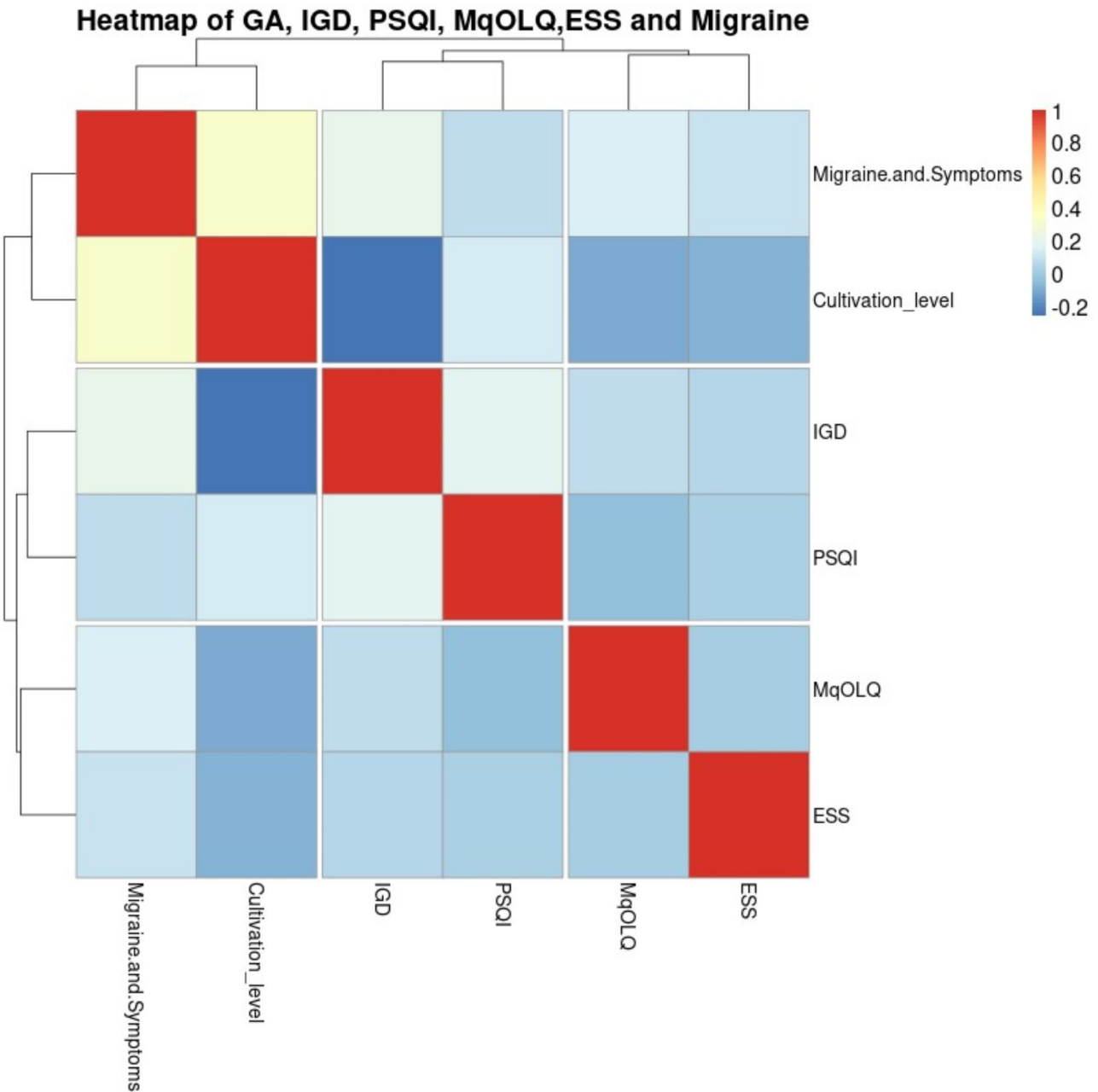


Fig. 1 Heatmap Correlation of Different Study Variables A correlation heatmap visually displays the strength and direction of relationships between variables, with correlation values r ranging from -1 to 1 calculated by Pearson's r correlation and significance has been measured by p -value

the model aims to uncover patterns and relationships that can provide deeper insights into how gaming behaviors and related factors contribute to migraine outcomes.

Parametric estimation of neural network model by MLP (Multilayer Perceptron Model)

The neural network model is designed to predict the likelihood of experiencing migraine pain or associated headache symptoms based on several input factors related to PUBG gaming and various health metrics. The input

layer consists of six factors: daily PUBG use (hours), gaming addiction, gaming disorder, Pittsburgh Sleep Quality Index (PSQI), 24-hour Migraine Quality of Life Questionnaire (24-hour MQoL), and the Epworth Sleepiness Scale (ESS). The complete data was divided into 80% training and 20% testing phase. 382 cases (79.9%) are used to train the MLP model while 96 cases (20.1%) are reserved for testing the model's performance. All 478 cases are considered valid and included in the analysis. 22 cases are excluded, leaving a total of 500 cases in the

dataset. The input layer has a total of 23 units (excluding the bias unit), which includes the original input variables transformed as necessary to fit the model. The output layer has one dependent variable, which is the likelihood of experiencing migraine pain or associated headache symptoms, represented by 2 units indicating the possible outcomes (experiencing migraine pain or not). The softmax activation function in the output layer converts the outputs into probabilities, ensuring that the sum of the probabilities equals 1. The cross-entropy function is used to calculate the error, which is a common practice in classification problems to measure the performance of a model whose output is a probability value between 0 and 1. This comprehensive approach allows for accurate prediction of migraine-related symptoms based on the specified input factors.

Figure 2 shows the Area under the Curve (AUC) values for the model predicting the likelihood of experiencing migraine pain or associated headache symptoms. The AUC is a performance measurement for classification models at various threshold settings, and it represents the ability of the model to distinguish between classes. An AUC of 0.690 indicates that there is a 69% chance that the model will be able to distinguish between a randomly chosen positive instance (experiencing migraine pain) and a randomly chosen negative instance (not

experiencing migraine pain). An AUC of 0.690, while not excellent, shows that the model has a fair ability to discriminate between those who experience migraine pain or associated headache symptoms and those who do not experience it while playing PUBG. The model's AUC suggests that the factors used in the input layer (daily PUBG use, gaming addiction, gaming disorder, PSQI, 24-hour MQoL, and ESS) provide some predictive power for determining the likelihood of experiencing migraine pain or associated headache symptoms.

The neural network model uses the normalized importance of various predictors to estimate the likelihood of experiencing migraine pain or associated symptoms [Table 3]. Normalized importance values help understand the relative significance of each predictor in the model. Daily PUBG use is the most significant predictor in the model, with the highest normalized importance of 100%. This means that the amount of time spent playing PUBG daily has the strongest influence on predicting whether an individual will experience migraine pain or associated symptoms. Gaming addiction is also an important predictor, but its impact is significantly lower compared to daily PUBG use. With a normalized importance of 36.9%, it contributes about 37% as much as daily PUBG use to the prediction of migraine pain or associated symptoms. Gaming disorder has a lower impact on

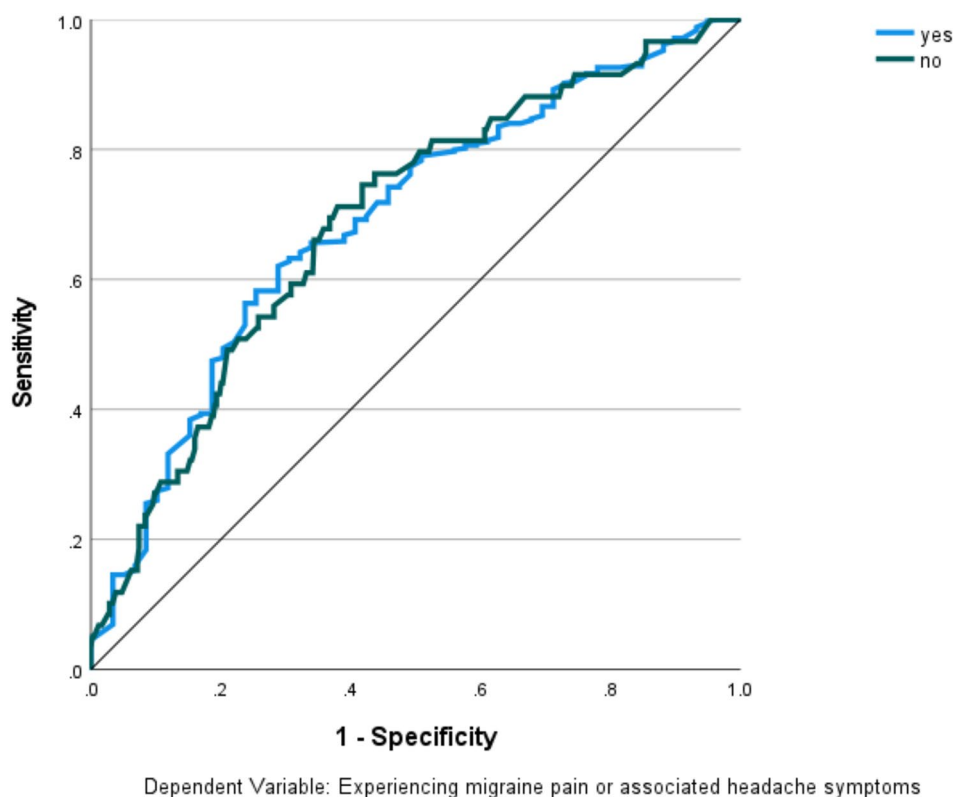


Fig. 2 AUC curve of estimating the Experiencing the Migraine and Associated symptoms False positive rate (1-specificity) has been plotted against True positive rate (Sensitivity) The internal offset = ± 0.5 . Interpretation of AUC is based upon Mann-Whitney U-test used for computing AUC curve

Table 3 Normalized importance of predictors in estimating Migraine and Associated symptoms

Independent Variables	Importance	Normalized Importance
Daily PUBG use (hours)	0.372	100.0%
Gaming addiction (GA Level)	0.137	36.9%
Gaming disorder (IGD Level)	0.070	18.8%
PSQI	0.191	51.3%
24-Mqol	0.109	29.3%
ESS	0.122	32.8%

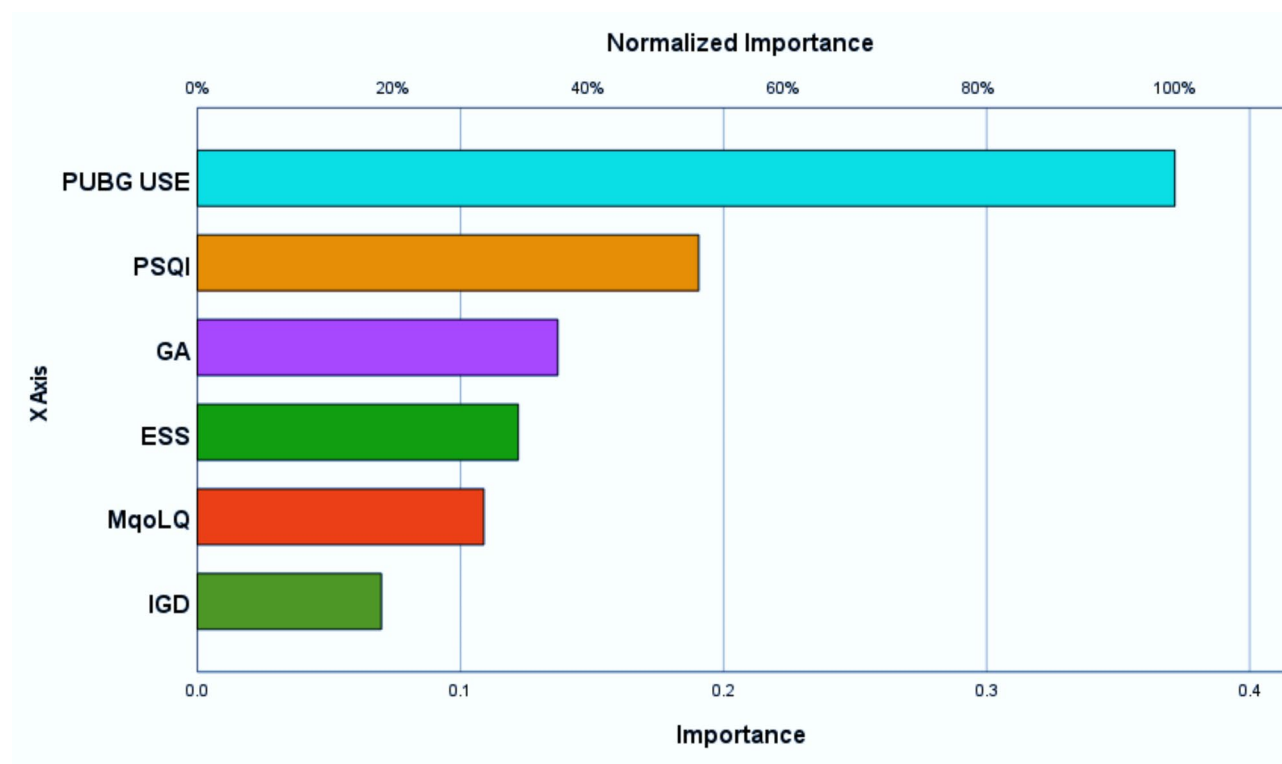
1. Normalized importance have been calculated for every predictor with one 1 hidden layer which includes 4 units using tangent hyperbolic activation function and 1 output layer with 3 units. 2. Softmax has been used as an activation function for output layer while cross-entropy has been computed for error-function

the prediction model, with a normalized importance of 18.8%. This indicates that while gaming disorder levels do play a role in predicting migraine symptoms, their influence is less than that of daily PUBG use and gaming addiction. PSQI is a significant predictor with a normalized importance of 51.3%, indicating that sleep quality is an important factor in estimating the likelihood of experiencing migraine pain or associated symptoms. Its contribution is more than half as influential as daily PUBG use. The 24-hour MQoL, which assesses the quality of life-related to migraine over 24 h, has a normalized importance of 29.3%. This shows that while it has a notable impact, it is less influential compared to daily PUBG use and PSQI. The ESS, which measures daytime

sleepiness, has a normalized importance of 32.8%. This indicates that daytime sleepiness is also a relevant predictor, but its impact is lower than that of daily PUBG use and PSQI [Fig. 3].

Discussion

This study investigated the relationship between various factors associated with PUBG usage and the likelihood of experiencing migraine pain or associated headache symptoms. Participants engage significantly with PUBG, the relationship between gameplay duration and the occurrence of migraine symptoms is weak. Despite a higher prevalence of migraines among more frequent players, the weak correlation between PUBG cultivation level and migraine pain indicates that factors beyond gameplay time likely contribute to the development of migraine symptoms. The relationship between PUBG use and migraine complaints, gaming addiction, sleep quality, and quality of life is minimal, indicating that PUBG cultivation has little impact on these variables. However, a moderate association with gaming disorder levels suggests that higher PUBG involvement may contribute to an increased risk of developing gaming disorders. Additionally, a small but noticeable connection exists between PUBG cultivation and daytime sleep disturbances, hinting at some influence on sleep patterns. These findings align with prior research which found that adolescents who watch TV for more than three hours daily are more

**Fig. 3** Percentage normalized importance of predictors in estimating migraine and associated symptoms

likely to report physical and psychological symptoms, including headaches [1]. Similarly, research on online gaming [22] has indicated a correlation between extended gaming sessions and tension-type headaches, consistent with our finding that 51.8% of HPU participants reported headaches and associated symptoms. Furthermore, studies [24, 25, 31, 32] examining smartphone and online game addiction among children and adolescents diagnosed with tension-type headaches or migraines have supported these observations. Results were also supported by the study in which it was observed that adult populations indicate a correlation between screen exposure and headaches [21]. This study did not reveal a significant association between PUBG gaming and overall sleep quality, as assessed by the Pittsburgh Sleep Quality Index (PSQI). However, we did observe an association with sleep disorders, as measured by the Epworth Sleepiness Scale (ESS). This supports existing findings that excessive screen time is linked to poor sleep quality and reduced quality of life [37]. The impact of PUBG on sleep quality may vary depending on individual gaming habits. Factors such as the duration and intensity of gameplay and the time of day at which gaming occurs could influence sleep patterns among participants differently. This suggests that further investigation into the specific effects of PUBG usage on sleep patterns and overall well-being is warranted, particularly concerning reduced sleep and poorer health outcomes [44].

Prolonged gaming is often associated with visual issues such as blurred vision, difficulty focusing, and symptoms of dry eye syndrome, along with the exacerbation of pre-existing conditions like myopia and hyperopia. Additionally, poor posture and extended periods of immobility during gaming can lead to physical discomfort, including neck pain and tension headaches. Our study found that higher levels of PUBG usage correlated with a lower reported quality of life, with 28.9% of high PUBG users expressing dissatisfaction. This finding aligns with previous research indicating that sleep cycle disruptions are common among esports players, as the activation of the sympathetic nervous system, particularly during evening hours, negatively impacts sleep patterns [45, 46].

While current findings do not fully support claims that excessive gaming universally leads to reduced participation in leisure activities, shorter sleep duration, and lower contentment with daily living [37], they do highlight notable variations in daytime sleep patterns among PUBG users. For example, 15.9% of high PUBG users reported higher-than-normal daytime sleep, suggesting potential disruptions in sleep-wake cycles that need further investigation.

The study rigorously examined the effects of PUBG gaming on migraine prevalence, sleep quality, and overall well-being. It revealed an association between high

levels of PUBG engagement and an increased frequency of migraines. Although a direct link between PUBG gaming and sleep quality was not established, the study did identify a notable correlation with sleep disorders among participants. Notably, machine learning analysis highlighted PUBG gaming as the primary contributing factor to migraine issues, followed by the PSQI, GA, and ESS. The results of this study can be used to highlight the potential health risks associated with excessive PUBG gaming, particularly its link to increased migraine frequency and associated sleep disorders computed by ESS.

Possible limitations of the study

The study's limitations include reliance on self-reported data, which may introduce some bias. The sample results may not be generalized for other populations and variations in gaming habits and potential confounding variables need a more detailed survey to study other confounding factors.

Acknowledgements

The authors would like to extend their appreciation to King Saud University for funding this work through the Researcher Supporting Project (RSP2025R481), King Saud University, Riyadh, Saudi Arabia.

Author contributions

Conceptualization, AH, MD, RRM, MA; methodology, AH, MD software, AH, MD, RRM, MA; validation, AH, MD, RRM, MA; formal analysis, AH, MD; investigation, AH, MD; data curation, AH, MD, DS; writing—original and revised draft and reviews, AH, MD, RRM, MA; supervision, AH, MD; project administration, AH; and funding acquisition. All authors have read and agreed to the published version of the manuscript.

Funding

The study was funded by the Researcher Supporting Project (RSP2025R481), King Saud University, Riyadh, Saudi Arabia.

Data availability

Data can be obtained by contacting the corresponding author if desired.

Declarations

Ethics approval and consent to participate

This study was approved by the Social and Health Welfare Department with Ethical Approval No.26/H-11-2021. Informed consent was obtained from each individual participating in the study on an electronic consent form.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

Received: 12 October 2024 / Accepted: 10 February 2025

Published online: 03 April 2025

References

1. Calderon ML. Implementing an Algorithm for Headache Management in a Patient-Centered Medical Home. 2021.
2. Olesen J. The International classification of Headache disorders: history and future perspectives. *Cephalalgia*. 2024;44(1):03331024231214731.

3. Bhattarai AM, Gurung S, Pathak BD, Karki S, Adhikari A, Tandon OP, Poudel S, Yadav D, Pant C, Dhakal B. Prevalence and clinical characteristics of headache among medical students of Nepal: a cross-sectional study. *PLoS ONE*. 2022;17:e0277821.
4. Langdon R, DiSabella MT. Pediatric headache: an overview. *Curr Probl Pediatr Adolesc Health Care*. 2017;47(3):44–65.
5. Tarantino S, Proietti Checchi M, Papetti L, Ursitti F, Sforza G, Ferilli MAN, et al. Interictal cognitive performance in children and adolescents with primary headache: a narrative review. *Front Neurol*. 2022;13:898626.
6. Monteith TS, Sprenger T. Tension type headache in adolescence and childhood: where are we now? *Curr Pain Headache Rep*. 2010;14:424–30.
7. Choi C, Hums MA, Bum C-H. Impact of the family environment on juvenile mental health: eSports online game addiction and delinquency. *Int J Environ Res Public Health*. 2018;15(12):2850.
8. Montagni I, Guichard E, Carpenet C, Tzourio C, Kurth T. Screen time exposure and reporting of headaches in young adults: a cross-sectional study. *Cephalalgia*. 2016;36(11):1020–7.
9. Palanichamy T, Sharma MK, Sahu M, Kanchana D. Influence of Esports on stress: a systematic review. *Ind Psychiatry J*. 2020;29(2):191–9.
10. Zieliński G, Pająk A, Wójcicki M. Global prevalence of Sleep Bruxism and Awake Bruxism in Pediatric and adult populations: a systematic review and Meta-analysis. *J Clin Med*. 2024;13(14):4259.
11. Romero-Reyes M, Bassiur JP. Temporomandibular disorders, Bruxism and headaches. *Neurol Clin*. 2024;42(2):573–84.
12. Ballabio M, Griffiths MD, Urbán R, Quartiroli A, Demetrovics Z, Király O. Do gaming motives mediate between psychiatric symptoms and problematic gaming? An empirical survey study. *Addict Res Theory*. 2017;25(5):397–408.
13. Nuyens FM, Kuss DJ, Lopez-Fernandez O, Griffiths MD. The empirical analysis of non-problematic video gaming and cognitive skills: a systematic review. *Int J Ment Health Addict*. 2019;17:389–414.
14. Brindova D, Veselska ZD, Klein D, Hamrik Z, Sigmundova D, van Dijk JP, et al. Is the association between screen-based behavior and health complaints among adolescents moderated by physical activity? *Int J Public Health*. 2015;60:139–45.
15. Hakala PT, Saarni LA, Punamäki R-L, Wallenius MA, Nygård C-H, Rimpelä AH. Musculoskeletal symptoms and computer use among Finnish adolescents: pain intensity and inconvenience to everyday life: a cross-sectional study. *BMC Musculoskelet Disord*. 2012;13:1–7.
16. Alqassim AY, Alharbi AA, Muaddi MA, Makeen AM, Shuayri WH, Safhi AM, Alfifa AY, Samily IH, Darbashi NA, Otayn MA, et al. Associations of Electronic device use and physical activity with headaches in Saudi Medical Students. *Medicina*. 2024;60:299.
17. Punamäki R-L, Wallenius M, Nygård C-H, Saarni L, Rimpelä A. Use of information and communication technology (ICT) and perceived health in adolescence: the role of sleeping habits and waking-time tiredness. *J Adolesc*. 2007;30(4):569–85.
18. Anderson CA, Bushman BJ. Effects of violent video games on aggressive behavior, aggressive cognition, aggressive affect, physiological arousal, and prosocial behavior: a meta-analytic review of the scientific literature. *Psychol Sci*. 2001;12(5):353–9.
19. Alexander LM, Currie C. Young people's computer use: implications for health education. *Health Educ*. 2004;104(4):254–61.
20. Blaauw BA, Dyb G, Hagen K, Holmen TL, Linde M, Wentzel-Larsen T, et al. The relationship of anxiety, depression and behavioral problems with recurrent headache in late adolescence—a Young-HUNT follow-up study. *Headache Pain Res*. 2015;16:1–7.
21. Özge A, Termine C, Antonaci F, Natriashvili S, Guidetti V, Wöber-Bingöl C. Overview of diagnosis and management of pediatric headache. Part I: diagnosis. *Headache Pain Res*. 2011;12:13–23.
22. Oksanen A, Metsähonkala L, Anttila P, Aromaa M, Jäppilä E, Viander S, et al. Leisure activities in adolescents with headache. *Acta Paediatr*. 2005;94(5):609–15.
23. Castro K, Rocket FC, Billo M, Oliveira GT, Klein LS, Parizotti CS, et al. Lifestyle, quality of life, nutritional status and headache in school-aged children. *NUTR HOSP*. 2013;28(5):1546–51.
24. Rocha-Filho PA, Santos PV. Headaches, quality of life, and academic performance in schoolchildren and adolescents. *Headache. J Head Face Pain*. 2014;54(7):1194–202.
25. Lima AS, Araújo RCd G, LRd MRAA, GFFd S, Cunha SB, et al. Prevalence of headache and its interference in the activities of daily living in female adolescent students. *Rev Paul Pediatr*. 2014;32:256–61.
26. Pawlowski C, Buckman C, Tumin D, Smith AW, Crotty J. National trends in pediatric headache and associated functional limitations. *Clin Pediatr*. 2019;58(14):1502–8.
27. Roth-Isigkeit A, Thyen U, Stöven H, Schwarzenberger J, Schmucker P. Pain among children and adolescents: restrictions in daily living and triggering factors. *Pediatrics*. 2005;115(2):e152–e62.Powers.
28. Patton SW, Hommel SR, Hershey KA. Quality of life in childhood migraines: clinical impact and comparison to other chronic illnesses. *Pediatrics*. 2003;112(1):e1–5.
29. Kowacs PA, Utiumi MA, Piovesan EJ. The visual system in migraine: from the bench side to the office. *Headache: J Head Face Pain*. 2015;55:84–98.
30. Rossi LN, Cortinovis I, Menegazzo L, Brunelli G, Bossi A, Macchi M. Classification criteria and distinction between migraine and tension-type headache in children. *Dev med Child Neurol*. 2001;43(1):45–51.
31. Neut D, Fily A, Cuvelier J-C, Vallée L. The prevalence of triggers in pediatric migraine: a questionnaire study in 102 children and adolescents. *J Headache Pain*. 2012;13:61–5.
32. Toyran M, Ozmert E, Yurdakök K. Television viewing and its effect on physical health of schoolage children. *Turk J Pediatr*. 2002;44(3):194–203.
33. Redmayne M, Smith E, Abramson MJ. The relationship between adolescents' well-being and their wireless phone use: a cross-sectional study. *Environ Health*. 2013;12:1–12.
34. Jain A, Jain P. Inter-relationship between gaming addiction, emotional intelligence, and psychological well-being of playerunknown's battlegrounds and non-playerunknown's battlegrounds online mobile game players: a comparative cross-sectional study. *Acta Med*. 2021;8(2):117–24.
35. Sawyer SM, Azzopardi PS, Wickremarathne D, Patton GC. The age of adolescence. *Lancet Child Adolesc Health*. 2018;2(3):223–8.
36. Mamun MA, Griffiths MD. The psychosocial impact of extreme gaming on Indian PUBG gamers: the case of PUBG (PlayerUnknown's battlegrounds). *Int J Ment Health Addict*. 2021;19(6):2170–4.
37. Aggarwal S, Saluja S, Gambhir V, Gupta S, Satia SPS. Predicting likelihood of psychological disorders in player unknown's battlegrounds (PUBG) players from Asian countries using supervised machine learning. *Addict Behav*. 2020;101:106132.
38. Pugalendhi R, Janet M. Impact study of PUBG game addiction on consumer culture of Chennai urban students. *Int J Sci Technol andManagement*. 2019;5(11):57–61.
39. D'Souza L, Manish S, Deeksha S. Development and validation of PUBG Addiction Test (PAT). *Int J Indian Psychol*. 2019;7(1):562–74.
40. Lemmens JS, Valkenburg PM, Peter J. Development and validation of a game addiction scale for adolescents. *Media Psychol*. 2009;12(1):77–95.
41. Izci B, Ardic S, Firat H, Sahin A, Altinors M, Karacan I. Reliability and validity studies of the Turkish version of the Epworth Sleepiness Scale. *Sleep Breath*. 2008;12:161–8.
42. Buysse DJ, Reynolds CF III, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry Res*. 1989;28(2):193–213.
43. Shen Y, Cicchella A. Health consequences of intensive E-Gaming: a systematic review. *Int J Environ Res Public Health*. 2023;20(3):1968.
44. Gerbner G. Cultivation analysis: an overview. *Mass Communication Soc*. 1998;1(3–4):175–94.
45. Video game disorder, Raouf SY, Gabr HM, Al-Wutayd O, Al-Batanony MA. And mental wellbeing among university students: a cross-sectional study. *Pan Afr Med J*. 2022;41:89.
46. Mylona I, Deres ES, Dere GS, Tsinopoulos I, Glynatsis M. The impact of internet and videogaming addiction on adolescent vision: a review of the literature. *Front Public Health*. 2020;8:63.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.